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Review article

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Tipping points in complex ecological systems

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Abstract

In the modern physics of complex systems, a topic receiving increasing attention is tipping points. These are usually thought of as states of a system in which changing parameters slightly can lead to a large, and generally abrupt and irreversible, change in properties. There exists a number of different mathematical structures that can show this behaviour, and the one that was originally used (nowadays often referred to as the bifurcation-induced tipping) is just one of many. Different preconditions and/or different levels of details included into the model, reflecting also different environmental forcing, can lead to various tipping mechanisms. Furthermore, in a spatially extended system and/or a system with multiple scales, different parts can react to a change in environmental conditions differently or at a different time, interacting with each other to create a tipping cascade. In this Review, using ecosystems as a paradigm of complex nonlinear open systems, we survey the progress made in tipping point science over the past 15 years and identify gaps in the knowledge.

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Key points

- Understanding, anticipating and predicting tipping points is a key research challenge for the study of ecological systems.
- The phenomenology of tipping points is complex and requires various mathematical techniques and ideas.
- The convergence of data-driven methods and theory-informed strategies is leading to encouraging progress.
- Tipping is increasingly seen as a complex phenomenon in which the emergence of spatial patterns can have a prominent role.
- Evidence for tipping in both empirical observations and in models is found in a number of important ecological systems.

Introduction

Over the past two decades, progress in studies on environmental systems (including climate and ecosystems across multiple scales), supported by the progress in understanding general properties of nonlinear dynamics and complex systems, has opened a new perspective on critical transitions by placing them into the context of dynamical systems theory. The system 'tips' or crosses a tipping point when its reference stable steady state either disappears[1,2] or loses it stability via a global bifurcation[3]. The most studied case is associated with the occurrence of a saddle-node bifurcation when a relevant parameter passes its bifurcation value (Fig. 1a). Once it happens, the system transitions to another steady state (or a stable dynamical regime, such as self-sustained oscillations); in the ecological context, such a regime shift is often associated with biodiversity loss and ecosystem degradation. This tipping caused by a parameter passing its bifurcation value is known as bifurcation tipping (or B-tipping).

Research in the past decade has considerably broadened the original understanding of tipping mechanisms. B-tipping is a paradigmatic scenario of abrupt changes in complex ecological systems[1,4]. However, it is not the only possible one. Indeed, the B-tipping mechanism apparently misses many factors that are typical in ecosystems, for example, the effect of stochastic forces and perturbations, and the existence of different timescales. A number of alternative dynamical mechanisms have been identified that can lead to ecosystems tipping (Table 1). As we discuss, the conditions required for them to occur differ substantially, and hence the mechanisms also differ in terms of their applicability and generality. Of course, nature does not follow theorems nor obey asymptotic limits, so any real-life example of tipping will in general have features belonging to more than one of the idealized dynamical mechanisms.

Questions may arise here: what is so special about ecosystems? Can they bring a new challenge to the science of tipping points that might not be present in other natural systems? As it happens, ecosystems indeed possess many dynamic and structural features that distinguish them from other natural systems, as is briefly outlined in Box 1. Many of these differences play out in canonical examples of tipping in ecosystems including coral reefs[5–7] and lakes[1,8]. Combinations of the above factors and an interplay between them make ecosystems highly complex systems, arguably more complex than any other natural system. This complexity has various implications, in particular it opens a possibility of tipping mechanisms different from classical B-tipping.

In this Review, we describe different types of tipping and how they apply in an ecological context. We then discuss cascading tipping, in which tipping in one subsystem can cause tipping in other subsystems. We then look at approaches to constructing early warning signals (EWSs), which are desirable for anticipating tipping in Earth systems. Finally, we discuss important case studies before pointing to some open questions in the field.

Types and scenarios of tipping

Bifurcation tipping

If one considers simple 1D models of the form $\dot{x} = f(x, v)$, in which v is a parameter, then B-tipping can also be understood in terms of a 'stability landscape' owing to a potential function. Because one can write $\dot{x} = f(x, v) = -\mathrm{d}V(x, v)/\mathrm{d}x$, the bifurcation described earlier occurs when a change in the parameter v leads to the elimination of the local minimum of V associated with the reference state (Fig. 1b). For high-dimensional, non-gradient systems (to which ecosystems belong), a potential function usually does not exist but tipping in response to a parameter change can be understood in a similar way using, instead, the notion of quasi-potential[9,10].

B-tipping includes some generic features of critical transitions such as the existence of a critical threshold, but it has much more to offer. B-tipping eliminates or destabilizes the present state, so the system has to transition to an alternative one. This is similar to a 'usual' critical transition in a physical system. However, one important difference (and not the only one) emerges from the fact that, in a certain range of environmental conditions (that is, for the same value of a control parameter such as temperature or precipitation), even in the non-spatial (spatially uniform) case, there may exist multiple stable states of the system. One of the best known examples is given by the savanna–forest bistability[11]; more examples will be given in the 'Case studies' section.

Although most visual representations of tipping are by necessity 1D, the scenario depicted in Fig. 1 can be generalized also for high-dimensional systems. If the system is described by a gradient dynamics, $\dot{x} = \mathbf{f}(\mathbf{x}, v) = -\nabla V(\mathbf{x}, v), \mathbf{x} \in \mathbb{R}^N$ in which $N > 1$, B-tipping occurs when, for a critical value of the parameter $v = h$, the local minimum merges with a saddle. Importantly, the scenario extends to the much more general and relevant case of non-gradient multistable systems, including those possessing competing states featuring chaotic dynamics. In this case, B-tipping occurs as a basin crisis[12] associated with the reference attractor touching an edge state[13,14] living in the basin boundary[15]. The edge state is a high-dimensional dynamical saddle embedded in the basin boundary, featuring one transversal unstable direction. As a result, although it is an invariant set, the edge state is only a relative attractor. It attracts trajectories initialized on the basin boundary; however, trajectories initialized in a neighbourhood of the edge state escape such a neighbourhood according to an exponential distribution of lifetimes[16]. If the dynamics on the edge state is chaotic, one might obtain the puzzling result that the basin boundary separating the competing stable states is fractal, having a co-dimension strictly smaller than one, and possibly very close to zero[14,15]. At the tipping point, the merging of the edge state and the reference attractor leads to the emergence of a ghost state[17–19].

The B-tipping scenario is a baseline which, strictly speaking, can only occur in (quasi-)autonomous systems in response to an infinitely slow parameter change. When the aforementioned analysis is applied

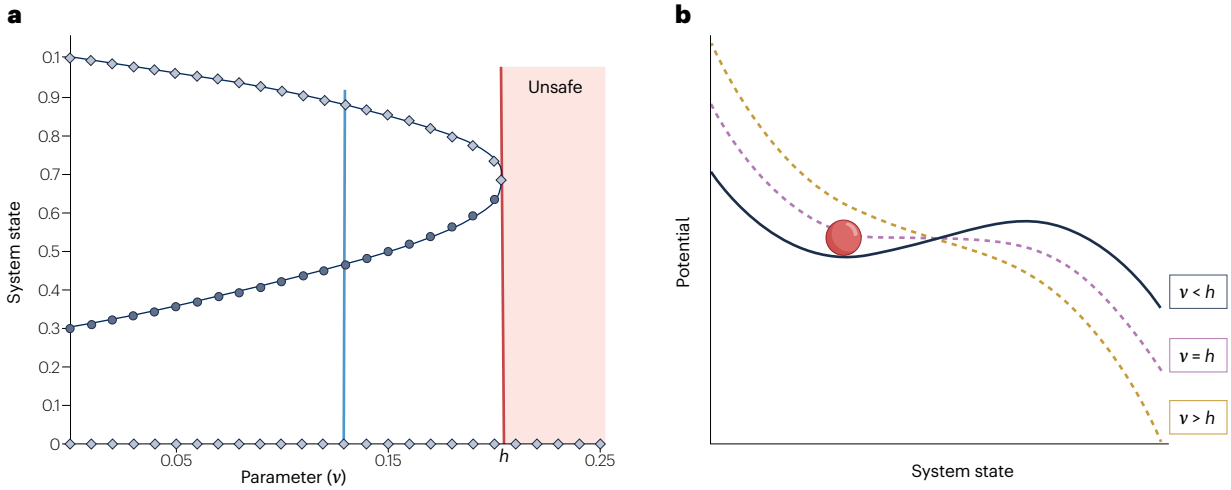


Fig. 1 | Tipping point in a prototypical 1D system with a few steady states. The system has the form $\dot{x} = -\mathrm{d}V(x, v)/\mathrm{d}x$, in which x is the state of the system, $V(x, v)$ is a potential function and v is the bifurcation parameter. **a**, A stable steady state of the system (shown by diamonds on the upper branch of the curve) can merge with an unstable state (shown by circles on lower branch of the curve) and disappear when a system's parameter v crosses its critical value at $v = h$ (shown by the red vertical line). The system is therefore in a 'safe' state for any $v < h$ (indicated by the blue line) but experiences a 'free fall' transition to another state for $v > h$ (indicated in red). **b**, The disappearance of the steady state in terms of $V(x, v)$: a sufficiently (overcritical) increase in v eliminates the local minimum, so that the system transitions to an alternative state (not shown). Panel **b** adapted with permission from ref. 27, CC BY-NC-ND 4.0.

to a real-world ecological or environmental system, a change in parameter values is usually associated with a change in the environmental conditions. Therefore, parameter v becomes a function of time, $v = v(t)$. In this case, the system becomes non-autonomous and scenarios of tipping become diverse and considerably more complicated. However, if the rate of change in the parameter value is slow compared with the intrinsic timescales of the system, the change in the properties of the system can still be analysed using a sequence of 'frozen' autonomous systems similar to the above. Another non-autonomous scenario, in which the concept of B-tipping can be applied, includes periodic forcing to mimic daily and seasonal variations of parameters in ecosystem models[20]. More rigorously, bifurcation theory for non-autonomous ordinary differential equations with periodic forcing can be used in this case[21]. Alternatively, one can sample the continuous flow at discrete time intervals equal to the forcing period and consider B-tipping of the corresponding autonomous time-discrete stroboscopic map.

It is possible to make a further step and introduce in this case a notion of quasi-potential[9,22,23], which generalizes the potential V mentioned earlier to the case of non-gradient systems. Practical construction of the scalar function, which summarizes the stability landscape of the system, requires adding stochastic forcing to the system, which are dealt with further in this Review.

Rate-induced tipping

The assumption underlying the B-tipping mechanism — that the controlling (bifurcation) parameter changes infinitesimally slowly — avoids the need to consider its explicit time dependence. Instead, one considers an ensemble of autonomous systems across different parameter values, extending to values at and beyond the bifurcation point. However, this assumption leaves open the question of how tipping would occur when the parameter changes too rapidly to be considered slow. The opposite limiting case, involving a rapid parameter change, is particularly relevant because the current rate of climate change is believed to be fast relative to the characteristic timescales of ecosystem resilience. Consider the case in which the value of the system parameter v changes from v_0 to $v_1 = v_0 + \Delta v$ over a certain time interval Δt, that is, with the average rate $R = |\Delta v/\Delta t|$. An ecological system can respond differently to perturbations of different rates[2,24–28]: it may be resilient to a slow change, so that its dynamics and properties remain largely unchanged, but if the change is sufficiently fast the system would experience a regime shift. This mechanism is commonly referred to as rate-induced tipping or R-tipping[2,29]. From the dynamical systems viewpoint, it can be thought of as a fast shift of the whole potential landscape, whereas the 'position' of the system remains approximately the same (Fig. 2a). As a result, the system may find itself outside its former basin of stability and consequently tip — that is, transition to an alternative stable steady state or regime — potentially causing drastic changes in ecosystem properties, such as species extinctions. Mathematically, the rate-induced tipping is described by the basin instability phenomenon[25] and the theory predicts tipping to occur for a supercritical rate of parameter change[25,30].

The details of the R-tipping scenario depend on the type of attractor on which the dynamical system resides before and/or after the tipping point. For instance, if the system transitions from a steady state to a steady state, there is a well-defined critical rate (say, R_{cr}) so that for any $R > R_{cr}$ the system leaves the previous attractor's basin (and hence a regime shift occurs). However, if the system state before or after tipping is more complicated than a steady state (periodic or chaotic, for example), the rate dependence becomes non-sharp and can even become non-monotonic. To quantify this change in the system's response — from a single critical threshold to a distribution of critical values — the concept of partial tipping has been introduced[31]. Interestingly, partial tipping can also be observed in a simple system in the case of a periodic forcing[32]. In the case of a chaotic attractor, the system attains probabilistic properties, so that the corresponding critical transition is better described by a distribution of tipping probabilities[33] rather than by a

Table 1 | Different types of tipping points and tipping mechanisms

Tipping type	Preconditions	Brief description	Examples
B-tipping (bifurcation tipping)	Change in parameter values (such as a persistent environmental change or trend)	Transition of a dynamical system to an alternative regime (such as a different steady state or oscillations with different frequencies or magnitudes)	Amazon forest[1], savanna vegetation[4]
R-tipping (rate-induced tipping)	Sufficiently fast (overcritical) rate of parameter change	Tipping that occurs if the rate of change in a certain parameter(s) exceeds a critical value	'Compost bomb'[191,192], r-tipping in bacterial communities, plankton blooms, complex mutualistic networks[2,25,26,28,193,194]
N-tipping (noise-induced tipping)	Presence of noise acting on state variables	Noise causing a system to transition to an alternative regime. Increasing the noise variance (noise strength) may increase the probability of tipping	Common in all natural systems with alternative steady states[195]
S-tipping (shock tipping)	Singular large perturbation of a state variable	Transition to an alternative steady state or regime following a single large perturbation of state variables	Extreme and/or rare weather events[56], bolide-Earth collision[55]
AN-tipping	Non-Gaussian, 'anomalous' noise (such as Levy-type)	Transition to an alternative steady state or regime, but with a distribution of exit times qualitatively different from N-tipping	Amazon forest[60], vegetation systems more generally[196], weather[197]
P-tipping (phase tipping)	Periodic or quasi-periodic dynamics	The probability of tipping (owing to the effect of noise or shock, for example) at certain specific times — 'phases' — is much higher owing to the geometry of the system's phase space. Increase in the noise variance may increase the likelihood of tipping	Cyclic ecosystems[65]
LT-tipping (tipping at the end of long transients)	Ghost attractors; saddle points; slow manifolds in systems with multiple timescales	Occurs in systems exhibiting a long transient: transient dynamics that mimics a self-sustained (asymptotic) regime but only lasts for a finite time	Inherent in many ecological and other environmental systems[66–68,129,198,199]

single deterministic value. A similar type of rate dependence can also be observed in stochastic dynamical systems[34]. Note that, in the case of a periodic or chaotic attractor, whether the system tips or not can depend on its initial position on the attractor. This dependence on the initial conditions may also lead to a distribution of tipping thresholds and hence can also be thought of as partial tipping[32,33].

Quantitative approaches for determining the critical rate of R-tipping extend beyond the classical bifurcation theory of autonomous systems. In particular, they use the concept of a (local) pullback attractor and examine bifurcations of such attractors as the rate of external forcing gradually increases. A pullback attractor extends the classical concept of an attractor in autonomous systems and can be understood as a time-dependent attracting set towards which a system evolves at time $t = t_0$, having started from specified initial conditions in the long past. Bifurcations of pullback attractors, occurring, for example, via collision with pullback repellors, provide the condition for R-tipping[35]. More elaborated methods deal with the scenario in which the time dependence of the parameter (external forcing) is asymptotically constant: it has finite limits when $t \to \pm\infty$, thus the external forcing remains globally bounded[36]. In this case, one can use the method of compactification[37]. The initial non-autonomous system is transformed into the so-called augmented system, in which time is considered to be an extra dynamical variable. The time is transformed in a way to make this additional variable bounded (mathematically, this can be done for exponentially asymptotically constant time dependence of the parameter). This gives an autonomous system with bounded solution. The obtained compactified system can then be analysed using the exiting methods for autonomous systems, for example, singular perturbation methods for fast–slow systems[36].

Other methods for predicting the critical rate in R-tipping include Melnikov-like methods (using asymptotic series expansions)[38], finite-time Lyapunov exponents[39], construction of safe regions around moving stable equilibria or various geometric singular perturbations methods[40,41]. However, existing methods have focused primarily on scenarios in which the moving states, or pullback attractors, are equilibria. Analytical methods for estimating the critical rate when the moving states are limit cycles or more complex attractors remain relatively underdeveloped[25]. Moreover, in ecosystem modelling, the assumption of asymptotically constant external forcing may not always be appropriate, as in the case of a gradual increase in global temperature. Therefore, new methods are needed to account for parameter changes that are locally bounded but not necessarily asymptotically constant[35].

In addition, in a non-autonomous system, the two generic tipping scenarios described earlier — tipping when a parameter crosses its critical value (B-tipping) and tipping when the rate of change exceeds a critical value (R-tipping) — can interact in a highly non-trivial way[42], leading to a more complicated structure of the parameter space and opening a possibility of tipping prevention. In particular, if the tipping point overshooting time is short compared with the intrinsic timescales of the system, tipping can be avoided through a parameter shift reversal[42,43].

Noise-induced tipping

The classical B-tipping mechanism does not require any stochasticity (which we refer to as noise). However, most natural systems do have noise and, under certain weak conditions about its properties, noise can occasionally drive a bistable or multistable system out of its current basin of stability and into the basin of another attractor (Fig. 2b). This phenomenon is known as noise-induced tipping or N-tipping. Note that this tipping scenario does not require any change in the system's parameters (meaning no change in the environmental conditions), because noise can act only on the state variables of the system. Correspondingly, because the effect of noise does not destroy the original stable steady state, a reverse transition may also be possible. The lifetime of the ecosystem in each of the stable steady states (or more generally, in each of the stable dynamical regimes) is therefore finite. On a longer timescale, the corresponding ecosystem dynamics becomes a

succession of alternating steady states (or regimes): the phenomenon that is sometimes referred to as metastability[44]. When proposing our classification of tipping scenarios, we should say that a strict borderline between R-tipping and N-tipping mechanisms is hardly possible, for example, in the case of highly correlated noise, as with the red noise[45]. Indeed, such a system would contain effects of stochasticity (N-tipping) and also depends on a variable parameter (R-tipping).

The notion of quasi-potential is an essential ingredient for performing a quantitative analysis of the metastability of a stochastically forced system. As a paradigmatic example, let us follow the Hasselmann programme[46,47] and consider the dynamics described by the following Ito stochastic differential equation[48]:

$$\mathrm{d}\mathbf{x} = \mathbf{f}(\mathbf{x}, \nu)\mathrm{d}t + \sigma\Sigma(\mathbf{x})\mathrm{d}\mathbf{W}, \quad (1)$$

in which $\mathbf{x} \in \mathbb{R}^N$, $\mathbf{f}: \mathbb{R}^N \to \mathbb{R}^N$ is a smooth drift, $\sigma > 0$ is the noise intensity, ν, as above, is a parameter, $\mathrm{d}\mathbf{W}$ is a vector whose components are N independent increments of Gaussian processes and $\Sigma(\mathbf{x}) : \mathbb{R}^{N\times N} \to \mathbb{R}^N$ defines the noise law. We assume that the noise is non-degenerate, which boils down to imposing that $\Sigma(\mathbf{x})\Sigma(\mathbf{x})^T$ is invertible everywhere.

Box 1 | Uniqueness of ecosystems as complex systems

Some of the distinctives of ecosystems are as follows. The list is not exhaustive and can be further extended[202,203].

Food web complexity. Prey–predator-type interactions along with interspecific competition bring different species together into a trophic network (that is, a food web)[204]. A typical food web consists of hundreds or even thousands of nodes (species) that are all unique and the network can have a complicated structure[93]. Disruption of food webs can lead to fast and substantial biodiversity loss on a regional[99] and even global scale[97], in the latter case potentially leading to mass extinctions.

Broadness of relevant spatial scales. Owing to the difference in the size of individual organisms (of species that are coupled together through trophic interactions, hence being parts of the same food web) and/or their typical dispersal distances, relevant spatial scales[205,206] can span across 10 orders of magnitude, ranging from 10^{-4} m (the size of microplankton in marine ecosystems, for example) to 10^{6} m (dispersal distance of some whales, birds, fish and so on).

Complexity of temporal dynamics. Complex intrinsic (self-organized) dynamics of interacting species, which includes periodic or quasi-periodic oscillations[207], deterministic chaos[208], intermittency[209], is further complicated by various external forcing[210–212] that can be deterministic and periodic (for instance, through diurnal or seasonal changes) or stochastic (oscillations in weather and climate, fluctuations in the solar activity or random anthropogenic interventions, for example).

Multiplicity of temporal scales. Complexity of temporal dynamics is further complicated by there being multiple timescales[213], which in ecosystems can have various origin[214]. In the simplest case, the existence of multiple timescales is provided by the differences in baseline timescales such as species average lifespan and the time between consequent generations. In an ecological community, these can span across a few orders of magnitude, ranging from hours (for instance, in seasonal phytoplankton reproduction) to decades (for some avian and reptile species).

Timescale of response. Phase transitions in physics are essentially changes in asymptotic behaviour[215]; by contrast, in ecology, the dynamics in the neighbourhood of a tipping point can play out slowly[7,214]. Thus, a system that has already tipped, and is on track to shift to a different state, can still be rescued by relatively rapid action. This has been reported for coral systems[7]. Timescales of response show up in other ways. The importance of global change for ecosystems which leads to nonautonomous systems means that rate-induced tipping is important[27,29].

Spatial complexity. The environment is heterogeneous on multiple scales. In terms of the species habitat or ecosystem type, this heterogeneity often leads to fragmentation when intrinsically continuous physical space appears to be split to a number of coupled patches, usually referred to as a metapopulation[216]. In turn, the fragmentation leads to new phenomena, for instance, a possibility of synchronization between population oscillations at different locations[217,218]. In a homogeneous environment, nonlinear interspecific interactions can result in self-organized pattern formation and spatiotemporal chaos[70,219].

Active species feedback on their environment. This can occur on different spatial scales, from local ('ecosystem engineers'[220] such as beavers) to global (changes in the global energy balance through changes in surface albedo[177,221]). In a slightly different context, the ability of species to amend their environment according to their needs is known as the Gaia hypothesis[222].

Species adaptive response and phenotypic plasticity. Contrary to abiotic components, biological species can change their traits and functions in response to changing environmental conditions[223]. Such phenotypic plasticity is generally believed to increase species fitness and hence can be expected to contribute to ecosystem resilience[224]. However, consequences of phenotypic evolution can be counter-intuitive, sometimes leading to a tipping point rather than mitigating it[225]. Also, the response can be taxon-specific and for some animal taxa may result in a strong Allee effect[226], which in turn may lead to species extinction[227]. A mismatch between the rate of environmental change and the rate of species adaptive response leads to ecosystem tipping, potentially resulting in mass extinctions[228].

Data challenges. Monitoring of ecological systems can be difficult compared with other types of system. Many methods proposed for early warning signs of impending tipping, for example, depend on data of a length and quality that are not available in ecological systems.

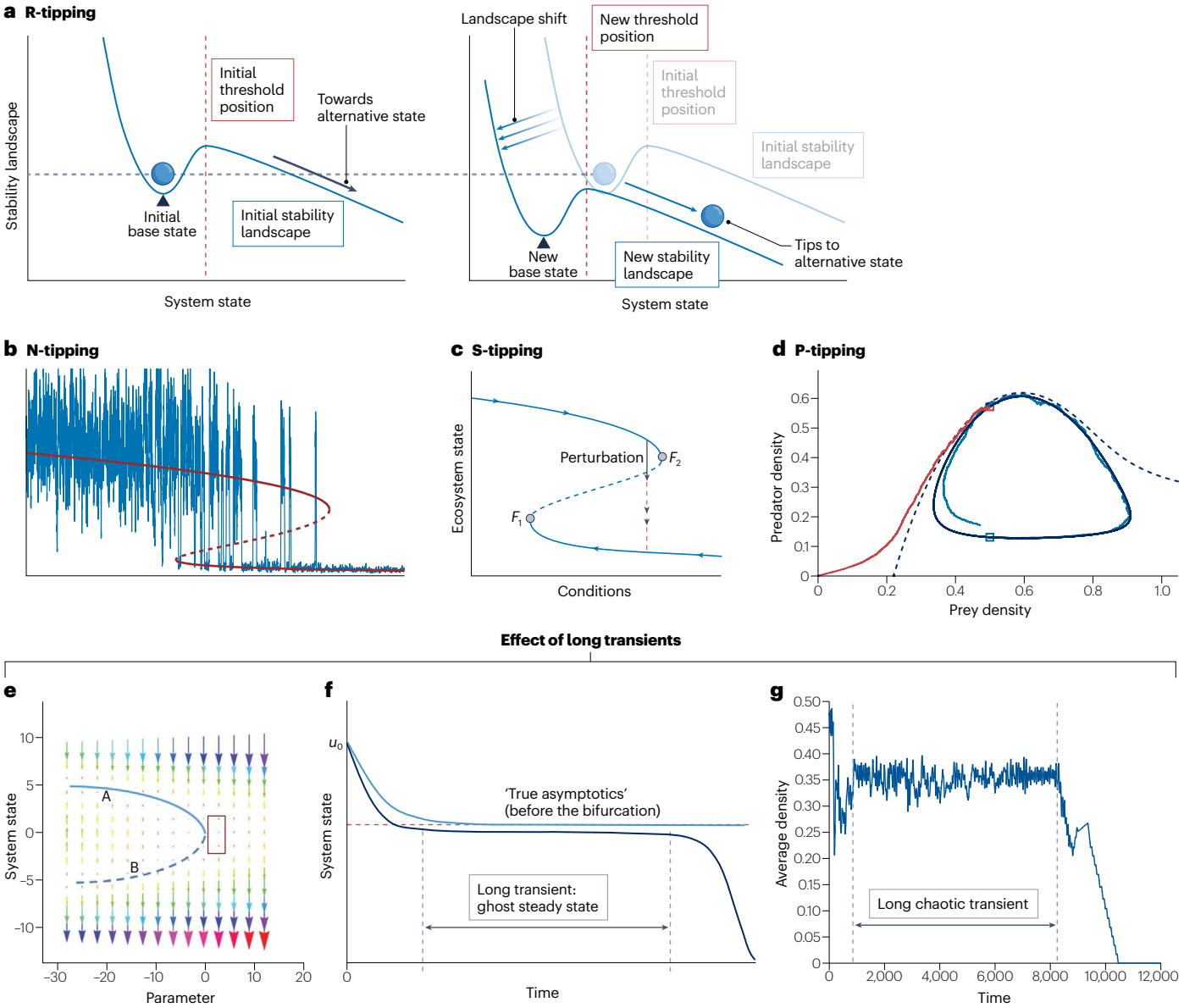


Fig. 2 | Different tipping mechanisms. a, Rate-induced tipping (R-tipping) occurs when a change in the environmental conditions occurs much faster than the system relaxation time. The system state remains approximately the same as before the change, but the whole quasi-potential landscape moves away[25,27]. **b**, Noise-induced tipping (N-tipping) occurs when a small stochastic perturbation (noise) occasionally pushes the system to an alternative steady state; the probability of such transition is larger if the system is closer to the saddle-node bifurcation. **c**, Shock-induced tipping (S-tipping) occurs when a single large perturbation is sufficient to push the system out of its previous basin of attraction (and hence to an alternative stable state or regime), even though it is not at a bifurcation point (indicated as F_1 and F_2)[1]. **d**, Phase-dependent tipping (P-tipping), shown here for a predator–prey system. If the underlying system dynamics is periodic, quasi-periodic or chaotic, the sensitivity of the system to perturbations may depend on its 'phase', that is, its position on the trajectory. The same noise may therefore lead (the red curve) or not lead (blue curve) to the system tipping, depending on the timing of the perturbation. The solid black curve shows the limit cycle and the dashed-and-dotted black curve shows the separatrix between the two basins of attraction. The red and blue squares show the position of the system in the phase plane when the perturbation was applied. **e**–**g**, Tipping due to the effect of long transients[66]. In the overcritical parameter range (shown by the red rectangle in panel **e**), there is no steady state but the dynamics is very slow. The slow dynamics may give an impression of a 'ghost steady state' (panel **f**). This phenomenon holds as well for more complicated dynamics, in particular in the form of a 'chaotic ghost'[68,199,200] (panel **g**). Panel **a** adapted with permission from ref. 25, CC BY 4.0. Panel **b** adapted with permission from ref. 195, AAAS. Panel **c** adapted with permission from ref. 1, Springer Nature Ltd. Panel **e** adapted with permission from ref. 201, National Academy of Sciences. Panel **g** adapted with permission from ref. 200, Elsevier.

The evolution of the probability density p of the system is described by the following Fokker–Planck equation[48]:

$$\frac{\partial p(\mathbf{x},t)}{\partial t} = \mathcal{L}_\nu p(\mathbf{x},t) \tag{2}$$

in which the operator $\mathcal{L}_\nu \bullet = -\partial_i(\mathbf{f}(\mathbf{x},\nu)_i \bullet) + \frac{\sigma^2}{2}\frac{\partial^2}{\partial_i \partial_j}(D_{ij}(\mathbf{x})\bullet)$ and $\boldsymbol{D} = \boldsymbol{\Sigma}\boldsymbol{\Sigma}^T$ is the diffusion coefficient. The invariant measure of the system $\rho(\mathbf{x},\nu)$ describes its steady-state statistics and obeys the following equation $\mathcal{L}_\nu \rho(\mathbf{x},\nu) = 0$. Under rather general hypotheses, in the weak noise limit $\sigma \to 0$, the invariant measure of the system can be approximated as $\rho(\mathbf{x},\nu) \sim Z(\mathbf{x},\nu)\exp(-2\Phi(\mathbf{x},\nu)/\sigma^2)$, in which $\Phi(\mathbf{x},\nu)$ is the so-called quasi-potential[9,22,23].

The function Φ is the generalization of the energy landscape for general nonequilibrium systems. Local minima of Φ are associated with attractors of the dynamics. If multiple competing states are present, Φ also provides information on the properties of the boundaries between the competing basins of attraction. Indeed, Φ is a saddle at the edge states. Note that if we consider smaller values of σ, the probability density concentrates in the local minima of Φ. Note also that if for a critical value of the parameter $\nu = h$ we have B-tipping, the attractor and the edge state collide and, similarly to the equilibrium case discussed above, the corresponding minimum and saddle of Φ merge. The accurate treatment of the simultaneous limits $\nu \to h$ and $\sigma \to 0$ is very nontrivial as B- and N-tipping processes interfere and gain relative prominence depending on how such limits are realized. In addition, the presence of noise can strongly interfere with the process of R-tipping by lowering the value of the critical rate of change of the parameter that is conducive to tipping[49]. Metastability can occur on multiple scales and can be organized in a hierarchical manner. This is then reflected in the properties of the quasi-potential Φ, as discussed in detail in ref. 44.

In the weak-noise limit, transitions between competing metastable states occur predominantly along special trajectories — the instantons — that connect an attractor and a nearby edge. Instantons can be obtained by solving a variational problem associated with the Friedlin–Wentzell action[50]. In the case of nonequilibrium systems, spontaneous relaxation trajectories (that is, heteroclinic orbits connecting an edge and an attractor) and instantonic trajectories are geometrically different[22,23]. Note that if we consider a weak but not infinitesimal noise, the most likely as well as the mean escape paths are better captured by solving a modified variational problem based on the Onsager–Machlup functional[51,52]. Finally, the average permanence time of the system in the vicinity of a reference state follows the scaling $\log(\tau) \sim (2\Delta\Phi/\sigma^2)$, where $\Delta\Phi$ is the lowest quasi-potential barrier confining the reference state[53].

Shock tipping

A dynamical system can also be pushed out of the basin of its stable equilibrium by a single short-term perturbation — a 'shock' — provided the magnitude is sufficiently large[54] (Fig. 2c). This mechanism is known as shock-tipping or S-tipping. Probably the most remarkable example of this tipping scenario occurred approximately 65 Ma ago when a large bolide collided with Earth; that singular event led to a dramatic global climate change, resulting in a mass extinction[55]. In the present-day Earth, S-tipping becomes increasingly relevant owing to the increased frequency and severity of extreme climate events[56]. In a more general case, there may be more than one shock. In that case, if the interval between the shocks is sufficiently small, that is smaller than a characteristic system's relaxation time, the effect of shocks can be mutually amplifying: although each individual one might be under-critical (not leading to tipping), a succession of them can make the system tip[57].

The timing and magnitude of shocks can usually be considered as random variables. One can therefore regard S-tipping as a particular case of N-tipping, in which the noise has special special properties. In a more general case, many small frequent random perturbations are combined with rare large ones. Remarkably, this is exactly the case of 'fat-tailed noise' described by a power law[58,59]. An important example of such heavy-tailed, impulsive processes is Lévy noise, which can be understood, in a coarse-grained sense, as a compound Poisson process in which singular perturbations occur at random times. Tipping induced by such 'anomalous noise' has properties that differ considerably from those of standard N-tipping driven by 'Gaussian noise'[60,61]. Indeed, in the case of fat-tailed noise, the notion of 'small noise' which is central to many studies becomes meaningless because the probability of a large perturbation occurring within any finite time interval is not negligible. For Lévy processes, the concept of a quasi-potential likewise becomes inapplicable. Instead, transitions occur when a singular perturbation displaces the system into the basin of attraction of a competing state, as defined in the absence of noise. To emphasize these qualitative differences, we refer in this Review to tipping induced by fat-tailed noise as AN-tipping. A more general theoretical framework for tipping induced by stochastic factors should consider a combination of Lévy and Gaussian noise[62–64].

Phase-dependent tipping

Many ecosystems and ecological communities exhibit cyclic dynamics — either self-organized, similar to the case of prey–predator cycles, or externally forced, by seasonal changes, for example. How sensitive such a system is to perturbations can depend on its position ('phase') in the cycle, and thus on when the perturbation occurs. For instance, noise of a given intensity may induce tipping when applied while the system trajectory is close to the boundary of its basin of attraction, but not at another point of time (Fig. 2d). Similarly a single perturbation of a given magnitude may induce tipping in one instance but not in another. This type of tipping due to a system's time-varying (phase-dependent) sensitivity is called P-tipping[65].

Long-transient tipping

Another tipping mechanism, which acts rather differently from the others described in this section, is what we refer to as long-transient tipping (or LT-tipping)[66]. A dynamical system can remain in a self-sustaining, apparently stable regime — mimicking an asymptotic state — for an extended period before suddenly undergoing a regime shift (Fig. 2e–g), but this tipping is not preceded by any change in parameter values or follows a parameter change that happened a very long time earlier. Long transients and the corresponding LT-tipping have been observed in many real-world ecological and environmental systems[67]. Note that because the existence of long transients is linked to common dynamical system structures (such as ghost attractors, unstable manifolds, multiple timescales and so on)[68], LT-tipping can happen in the absence of noise or shocks.

Tipping in spatially explicit systems

Perhaps one of the biggest challenges is understanding tipping in spatially explicit systems. The tipping mechanisms outlined earlier (see also Table 1) have been discovered and studied predominantly in non-spatial models. Models with explicit space can have different properties and, overall, much richer dynamics, acquiring features that

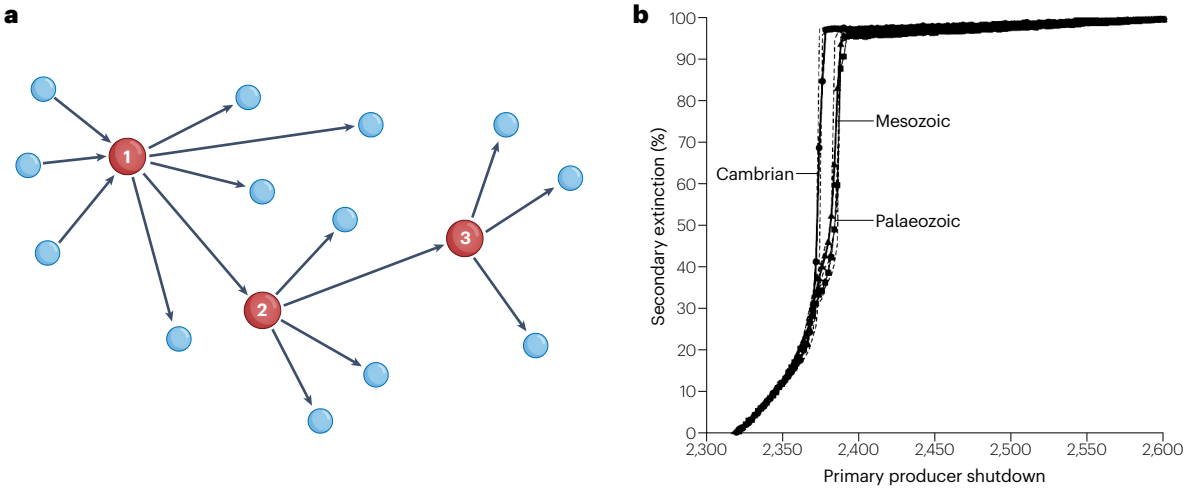


Fig. 3 | Cascading tipping on trophic networks. a, An example of heterogeneous food web in which tipping (species extinctions) in bottleneck subsystems (nodes 1–3) leads to tipping in other parts of the web. **b**, Effect of tipping cascades in an ancient food web[97] resulting in a highly nonlinear, threshold-type response of the ecosystem to the initial tipping in parts of the web. Panel **b** adapted with permission from ref. 97, Cambridge Univ. Press.

simply do not exist in their non-spatial counterparts. There is some theoretical and empirical evidence for spatial systems tending to be more resilient to an environmental change. The increased resilience can arise from various mechanisms[69], in particular from self-organized pattern formation[24,70,71]. This effect can substantially alter tipping scenarios[72], sometimes to the extent that an abrupt response to a change in environmental factor (parameter) becomes gradual[73,74]. There can be novel tipping mechanisms specific for spatially extended ecosystems. For example, in a fragmented habitat, global tipping can occur when the strength of diffusive coupling between local populations exceeds a critical threshold[75]. Also, tipping points can be determined by conditions leading to synchronization between the population dynamics at different locations. In the case of large-magnitude population oscillations in which the population density falls to a low level, synchronization can lead to tipping by turning a local population extinction to the global one[76]. In a spatially heterogeneous biome, the response to a global environmental change can lead to a fragmented tipping: coexisting states with part of the biome in one state and part in another[77]. A similar phenomenon is known as clustered tipping[78]. Finally, in a spatially explicit non-autonomous system that accounts for habitat movement (for example, towards the poles, to escape the effect of global warming[79,80]), R-tipping – species extinction – occurs when the movement rate exceeds a critical value[81]. These findings provide a quantitative framework for assessing the vulnerability of populations to rapid habitat shifts[82].

Rethinking of tipping points

Progress in the science of tipping points has led to a broadening of the definition of what a tipping point is and/or how system's tipping actually occurs. Recall that, according to the original understanding of tipping phenomena[2,83], in the absence of noise, a necessary condition for a system to tip is the existence of a well-defined critical threshold (in a bifurcation parameter for B-tipping or in the rate of change for R-tipping or in the magnitude of the shock for S-tipping). However, it becomes somewhat different if the system state before or after tipping is more complicated than a steady state, for example, if it is periodic (limit cycle) or chaotic (strange attractor). In that case, there is no single threshold because the system's response to perturbations may depend on its position on the attractor. Instead, tipping is described by a distribution of critical thresholds, which is captured by the concept of partial tipping[31–34].

In the presence of noise, a critical threshold is not required. Moreover, in the list of tipping mechanisms (Table 1), a threshold is not a necessary condition for tipping to occur even in the absence of noise. In particular, if tipping occurs at the end of long transient dynamics (LT-tipping) – due to a ghost attractor[66], the existence of multiple scales (through slow–fast dynamics[68], for example) or the effect of saddle points[84] (including chaotic saddles[12,85–87]) – a threshold is not needed.

Cascading tipping

Owing to its inherent structural and spatial complexity, an ecosystem can often be considered to be a number of subsystems coupled among themselves. This leads to a possibility of cascading tipping (or a tipping cascade) when tipping in one subsystem, such as a species extinction, triggers tipping in other subsystems through a sort of chain reaction[88–90]. This can happen in several different contexts. One of them is the heterogeneity of food webs and the existence of bottlenecks and keynote species. Tipping in any of the bottleneck subsystems (Fig. 3a) would likely lead to tipping in other connected subsystems[91,92]; this is called secondary extinction (Fig. 3b). The extent of such a 'domino effect' and/or the magnitude of the resulting impact on the ecosystem as a whole depends on the position of the bottleneck in the food web[93]. Extinction of primary producers known as 'bottom-up' cascade would intuitively lead to tipping (or extinction) in many other subsystems at higher trophic levels. Similar cascading effects have also been reported in physical and engineered systems, including power grid failures in which collapsing of the first element can result in subsequent numerous failures of other connected nodes[94]. Much less intuitively, owing to highly nonlinear interactions between species in a food web, a 'top-down' cascade[95] can also occur[96,97]. Tipping at a high trophic level (extinction of a top predator, for example) may lead to tipping at low trophic levels of the food web by releasing pressure on

intermediate consumers and hence causing an over-exploitation of a food resource[96,97].

Cascading tipping in ecosystems may occur on networks of a different origin. Owing to its spatial structural complexity, an ecosystem often consists of a fragmented habitat in which the same species exists at different separated spatial locations coupled by dispersal. Species extinction in one location or a collapse of one local community may then lead to a collapse of the entire dispersal network and thus the whole ecosystem[98,99].

In Earth systems more generally, in which ecosystems are coupled to climate[83,100,101] and to social and economic systems[102,103], tipping cascades may follow various even more complicated scenarios. We first consider cascades in fully deterministic systems. One tipping scenario occurs in leader–follower systems (also known in the literature as 'master–slave' or 'driver-driven' systems) in which the leader subsystem exerts control over the follower subsystem, which in turn can regulate another follower system[88,104]. We assume that each isolated subsystem (a tipping element) within this unidirectional chain has at least two qualitatively different coexisting states. If each subsystem in the chain is close to its tipping threshold, tipping in the leader system would propagate down the chain resulting in a domino effect[89,90]. An example of such cascade tipping via the leader–follower mechanism is the observed regime shift between eutrophic and oligotrophic states in a connected cascade of freshwater lakes[105]. Interestingly, even in a simple unidirectional chain, the phenomenon of cascading hopping can occur, in which propagation of tipping across the system may hop over some subsystems if they are characterized by higher tipping thresholds. Such subsystems do not tip themselves yet trigger tipping in their neighbour systems down the chain[88,104].

In a leader–follower chain, another phenomenon is indirect tipping[88], in which tipping does not occur in the leader system that is subject to the parameter drift but instead takes place in the follower subsystem. In a leader–follower chain with three subsystems, more complex combinations of cascading tipping and indirect tipping can occur, as the parametric drift occurring in the leader subsystem can cause tipping in either of the two follower systems down the chain[88]. A more complicated scenario involves bidirectional connections, in which subsystems affect each other via positive or negative coupling[90].

Interacting subsystems are often connected via a complex network and the nature of this network has a crucial role in the occurrence and propagation of tipping cascades[106]. The domino effect on a complex network has been reported in a study on the Amazon rainforest degradation, in which coupling between local patches is due to moisture-recycling mechanisms[107]. Another empirical study shows a domino effect in a large spatial aquatic system in the western Baltic Sea. Propagation of a regime shift through space was observed from the open sea towards the mainland, resulting in a change of the dominance from large predatory fish to smaller prey fish[108]. Importantly, the realization of tipping cascade on a network can depend on which subsystem has initially tipped[109,110].

We emphasize that tipping in one subsystem (as a result of an increase in external forcing, for example) is not by itself sufficient to initiate a tipping cascade, particularly if the coupling between the subsystems is weak. However, if an increase in the forcing continues, it may cause other subsystems tip before their own tipping points are crossed: that is, earlier than would happen for each subsystem in isolation. In such a case, the change of the internal state in the subsystem that tips first effectively works as an addition to the external forcing. Note that, depending on the rate of change in the external forcing, the interval between the tipping of the first subsystem and tipping of other subsystems can be long. This is called a two-phase tipping[90] or a multiphase tipping in case of multiple tipping elements. A well-known example of two-phase cascading in a climate system is given by the interaction between the Atlantic Meridional Overturning Circulation (AMOC) and the El Niño/Southern Oscillation (ENSO)[90,111]. An example of multiphase cascading in a complex socio-economic-environmental system that included 12 countries is given by the Arab spring (2010–2011). In this system, sudden massive street protests (understood as tipping[104]) started locally but then propagated across a large geographical region triggered by a change in some global factors (such as rising food prices across the entire region). However, the timing of the protests was not synchronized, and in some 'semi-isolated' countries (that were only weakly coupled to the rest of the system) it did not start until after a substantial delay[104]. Finally, we mentioned that tipping cascade can also be triggered if the parameter change is fast enough; this is known as a rate-induced tipping cascade[112].

A key aspect of multiphase cascades is the characteristic timescales of tipping elements[113]. As a prototypical example, consider a unidirectional leader–follower chain containing interconnected subsystems A, B, C and so on. Let the state before tipping be $(A, B, C, \ldots)$. An initial collapse of A can trigger a domino effect causing the subsequent collapse of B, which in turn will result in the collapse of C, eventually leading to the final 'tipped' state $(A^*, B^*, C^*, \ldots)$. If the internal characteristic timescale of subsystem B is long, after subsystem A tips, subsystem B remains in its original state for some (possibly very long) time, so that a transient 'metastable' state $(A^*, B, C, \ldots)$ emerges[114]. The lifespan of the metastable state(s) can be made longer by the presence of long transients caused by ghost attractors and crawl-bys (saddles)[66,67]. However, if the internal timescales of subsystems B and C are short, all three subsystems tip at approximately the same time, so that a fast transition to the final state $(A^*, B^*, C^*, \ldots)$ occurs. This scenario is called joint cascading[90] (also known as a fast or synchronized domino effect[115]). The difference between domino cascading and joint cascading is schematically shown in Fig. 4a,b. An empirical example of joint cascading is arguably the reported degradation of the Caribbean coral reefs with a quick loss of about 50% of the coral cover[116].

Another tipping cascade may occur in a chain or network of subsystems distributed along an environmental gradient, with their tipping thresholds arranged hierarchically along that gradient. In this scenario, an increase in global external forcing, such as rising temperature, could trigger a sequence of tipping events across the chain, resembling a domino effect. However, an important difference is that, in this scenario – which we term quasi-cascading – tipping in neighbouring subsystems is not necessarily caused by direct interactions. Indeed, it can occur in systems comprising uncoupled elements or elements with only weak coupling[111].

The inclusion of environmental and demographic stochasticity can modify the scenarios described earlier and alter the classification of tipping cascades. Noise may trigger tipping cascades that would not occur in the corresponding deterministic system. In the presence of noise, multiphase cascading and domino effects may also become difficult to distinguish. In a slow domino effect, noise-induced tipping can cause a regime shift in one subsystem, whereas the remaining subsystems persist in their current states for an extended transient period[115,117]. However, noise can reduce the transient time of the domino effect when compared with the corresponding deterministic system[115]. In particular, a spatially correlated noise can synchronize the dynamics of separate tipping elements, hence turning the domino effect into joint cascading.

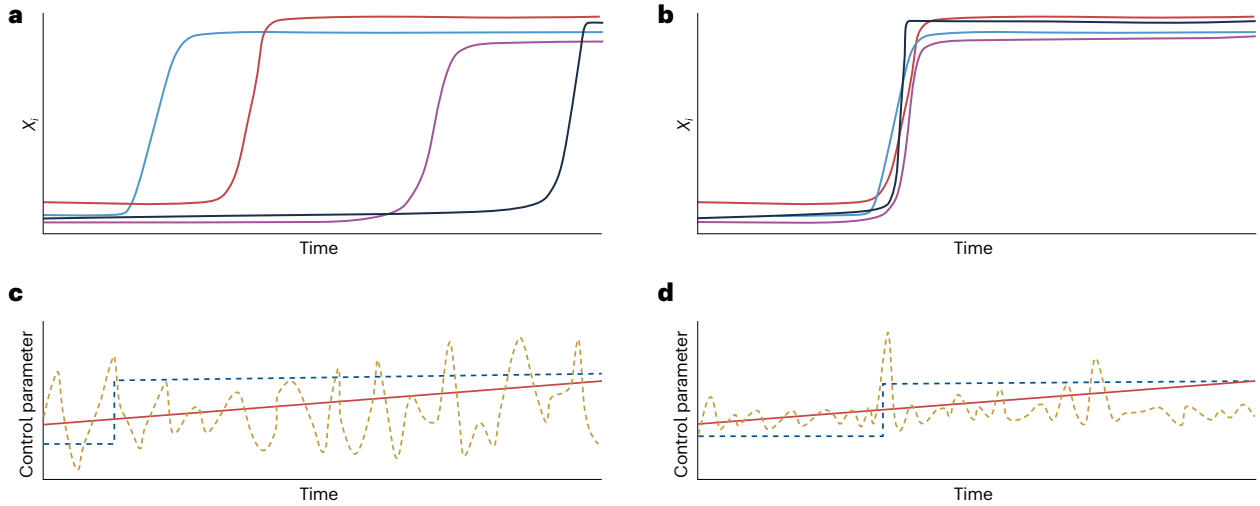


Fig. 4 | Different types of cascading tipping. a, Dynamical pattern of cascading tipping in a system of coupled bi-stable subsystems $X_i(t)$ on a network. Initially, all subsystems stay in the lower (non-tipped) state; further tipping of each subsystem occurs as a separate event. **b**, Joint cascading tipping (the joint tipping or fast domino effect): regime shifts of X_i occur simultaneously. In both types of cascading tipping, the mechanism behind the cascading can be different, depending on how the control parameter varies, as shown in the bottom panels. **c**,**d**, Scenarios of variation of the control parameter that result in the cascading shown in panels **a** and **b**, respectively: a gradual increase (solid line), stochastic variation (dashed line) and step-wise abrupt switch (dashed-and-dotted line).

The existing mathematical theory of tipping cascades is largely limited to networks of bistable elements, often comprising only two or three subsystems connected by unidirectional coupling[90,115]. This is because of the complexity of the system. Adding just one extra tipping element to the network or considering more complicated coupling (such as a bump-shaped coupling, or non-local and/or nonlinear coupling with its strength dependent on the state of the adjacent subsystems) may bring a new type of tipping[88,114]. Modelling cascading effects in real-world Earth systems that include multiple interacting biological and non-biological subsystems is generally extremely challenging[101]. A typical approach consists of constructing a 'master model' that includes a large number of subsystems. The model is then studied via extensive simulations trying to connect potential tipping events in individual subsystems. However, such a 'brute force' approach is often lacking in understanding of the generic mechanisms behind the tipping[101,118]. For example, it may be difficult to conclude which particular type of cascading scenario is realized[89] (Fig. 4).

Early warning signals

Anticipating tipping points caused by various mechanisms is of key importance in Earth sciences, and the literature on EWSs of critical transitions is abundant. In particular, many systems show similar behaviour when they approach a critical transition: their variance and autocorrelation increase[119]. Most existing methods of EWS are summarized in a 2024 review[120]. The classical methods of EWS are based on the concept of critical slowing down (CSD): near a bifurcation point, the recovery rate of the system progressively deteriorates; therefore, to detect possible tipping point, one needs to follow the key indicators, such as the variance, temporal and/or spatial correlation and recovery rate. However, classical CSD-based indicators of regime shift often fail to provide reliable forecasts. For example, they are unable to reveal the regime shift even while an abrupt transition occurs in the system[120,121]. Another possible reason for the failure of the CSD-based methods is the effect of noise. Indeed, in the case in which the confining quasi-potential (Fig. 1) is shallow, the destabilizing effect of stochasticity may be not compensated by the system's attraction to the stable steady state, so that even the slightest amount of noise (or of extra active forcing) would cause the system to tip[122].

It is important to note that although CSD is conceptually linked to bifurcation-induced transitions, in actual systems the EWS can mimic CSD without being caused by an approaching tipping point[123]. As an example, both variance and autocorrelation may increase in purely linear systems simply due to non-stationary correlated noise[124].

Applying the formalism of the Koopman–Kolmogorov operator[125] or, equivalently, of the Perron–Frobenius operator[126], it is possible to greatly generalize[127,128] the classical CSD arguments for systems whose dynamics is described by equation (1). The two approaches are equivalent, because the first focuses on the evolution of observables and the second on the evolution of probability measures. We recall here some key results; a more detailed analysis can be found in refs. 47,64,127. Under rather general conditions, the operator $\mathcal{L}_v$ introduced in equation (2) admits pairs of eigenvalues and eigenvectors (λ_j, ρ_j) such that $\mathcal{L}_v\rho_j = \lambda_j\rho_j$. The eigenvalues λ_j have non-positive real part, in which $\lambda_0 = 0$ is the eigenvalue corresponding to the invariant measure, as discussed earlier. The complex-valued λ_j's can be hierarchically ordered as $\lambda_0 = 0 > \mathrm{Re}(\lambda_1) \geq \mathrm{Re}(\lambda_2) \geq \ldots$ and represent intrinsic relaxation rates of the ensembles shaping the statistical dynamical properties of the system associated with the Perron–Frobenius modes ρ_j's.

Equivalently, one can instead study the eigenpairs (λ_j, ϕ_j) such that $\mathcal{L}_v^T\phi_j = \lambda_j\phi_j$, in which $\mathcal{L}_v^T$ is the adjoint of the operator $\mathcal{L}_v$. In this case, the ϕ_j's are the Koopman–Kolmogorov modes.

The discrete, isolated, eigenvalues $\{\lambda_j\}_{j=1}^N$, with N possibly infinite, are sometimes called (stochastic) Ruelle–Pollicott resonances[127]. The so-called continuous spectrum describes the very fast decaying modes. Figure 5 provides a schematic depiction of the spectral properties of $\mathcal{L}_v$.

It is possible to approximate the lagged autocorrelation of a general observable Ψ as[47]:

$$C_{\psi,\psi}(t) = \sum_{j=1}^{N} \alpha_j(\Psi) e^{\lambda_j t}, \quad (3)$$

in which the formula for the coefficients $\alpha_j(\Psi)$ can be found in ref. 47. The essential result is that no matter the choice of the observable, the correlation is written in terms of dictionaries of given exponential functions[47,128].

Tipping behaviour occurring as $v \to h$ is associated with the real part of the subdominant Ruelle–Pollicott eigenvalue approaching zero (or, in other terms, the spectral gap $\tau = |\mathrm{Re}(\lambda_1)|$ shrinking to zero), so that the corresponding characteristic timescale of decay diverges (Fig. 5). If τ shrinks for a pair of eigenvalues, then at the transition there is a critical oscillation, which is reminiscent of a Hopf bifurcation but occurring in potentially very high dimension[127,129]. The mode ρ_1 associated with the subdominant λ_1 (or with the pairs of critical complex conjugate λ's) defines the critical mode of variability, which can be identified with the so-called degenerate fingerprinting[130]. Such a mode defines the precursor of the critical transition. The lagged correlation $C_{\psi,\psi}(t)$ can act as an EWS for the tipping behaviour only if the chosen observable has a strong projection on the critical mode. This becomes especially clear if one considers the expression for the integrated correlation time for

$$\int_0^\infty \mathrm{d}\tau \tau C_{\psi,\psi}(\tau) = \sum_{j=1}^{N} \frac{\alpha_j(\Psi)}{\lambda_j}. \quad (4)$$

Hence, if one considers systems that are even moderately dimensional, a well-informed choice of the observable is critical for defining an efficient EWS. Observables that clearly distinguish between the attractor and the nearby edge state tend to have a very strong projection on such a critical mode, thus providing a targeted early-warning indicator[131].

The principal advantage of the Kolmogorov-operator formalism is that it enables a fully data-driven analysis of the system of interest. This is particularly valuable given the efficiency and power of variants of extended dynamic mode decomposition[132,133], which can be used to find to determine the operator's eigenpairs with high accuracy and computational efficiency (ϕ_j, λ_j). Although the standard method suffers from the curse of dimensionality, its kernelized version[134] is adaptive and seems well suited for studying the spectral properties of high-dimensional datasets. The first attempts in this direction are proving encouraging[135].

The Koopman framework is complemented by deep-learning-based algorithms, which offer an alternative to traditional CSD indicators. Deep-learning methods can handle systems with multiple interacting timescales without requiring explicit prior knowledge of the underlying equations. Moreover, they can identify the specific dynamical modes or spatial patterns that contribute most to impending tipping behaviour[136,137]. This ability to perform both detection and attribution makes deep learning particularly attractive. Furthermore, deep learning methods show promising results for anticipating R-tipping behaviour[138]. This finding suggests that deep learning may help bridge the gap between classical CSD theory — which often assumes slow or quasi-static forcing — and real-world scenarios in which forcing rates may be rapid and time-varying.

Note that cascading tipping poses additional challenges, because the autocorrelation of two time series and their interaction need to be analysed simultaneously. An altered form of de-trended fluctuation analysis — dubbed de-trended cross-correlation analysis — has been used to assess the cross-correlation between two non-stationary time series[139]. In this approach, the fluctuation function is computed using the cross-covariance rather than auto-covariance and is fitted to a power law. EWS in the context of indirect tipping poses additional challenges because studying the correlation properties of the leader system brings no signature of the impeding tipping in the follower system[88].

The functional analytical framework discussed earlier is extremely useful for clarifying EWS in the case of cascading tipping and indirect tipping. Assume the following leader–follower form for equation (1):

$$\begin{aligned} \mathrm{d}\mathbf{x}_1 &= \mathbf{f}_1(\mathbf{x}_1, v)\mathrm{d}t + \sigma_1 \Sigma_1(\mathbf{x}_1)\mathrm{d}\mathbf{W}_1 \\ \mathrm{d}\mathbf{x}_2 &= \mathbf{f}_2(\mathbf{x}_1, \mathbf{x}_2)\mathrm{d}t + \sigma_2 \Sigma_2(\mathbf{x}_2)\mathrm{d}\mathbf{W}_2, \end{aligned} \quad (5)$$

in which $\mathbf{x}_1 \in \mathbb{R}^M$, $\mathbf{f}_1 : \mathbb{R}^{M+1} \to \in \mathbb{R}^M$, $\mathbf{x}_2 \in \mathbb{R}^{N-M}$, $\mathbf{f}_2 : \mathbb{R}^{N-M} \to \mathbb{R}^{N-M}$, $\sigma_1, \sigma_2 > 0$ modulate the noise intensity, $\mathrm{d}\mathbf{W}_1$ and $\mathrm{d}\mathbf{W}_2$ are vectors whose components are M and $N-M$ independent increments of Gaussian processes, respectively, and $\Sigma_1(\mathbf{x}_1)\Sigma_1(\mathbf{x}_1)^T$ and $\Sigma_2(\mathbf{x}_2)\Sigma_2(\mathbf{x}_2)^T$ are non-negative definite matrices. Finally, v is the parameter of the leader system we alter to explore critical behaviour. In this case, the eigenvalue set expands beyond the leader's spectrum, owing to the follower's internal dynamics. The eigenfunctions depend in general on both $\mathbf{x}_1$ and $\mathbf{x}_2$, so no obvious decoupling is realized.

In the case of cascading tipping, one considers a situation in which, as v approaches its critical value, the leading system approaches a tipping transition. In this case, it is possible to find a suitable observable $\Psi = \Psi(\mathbf{x}_1)$ (or, more generically, $\Psi = \Psi(\mathbf{x}_1, \mathbf{x}_2)$) whose lagged correlation can act as an efficient EWS. In the case of indirect tipping, criticality is associated with the dynamics of the follower system. As a result, the projection of the critical mode associated with $v \to h$ on an observable that depends only on $\mathbf{x}_1$ will be vanishing. Hence, in practical terms, no EWS can be achieved by looking at the leader system only. In many applications, this implies the need for a network-aware and/or system-wide monitoring.

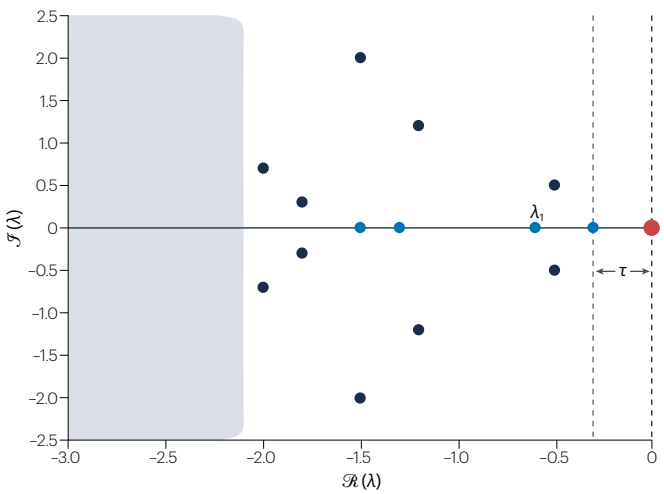


Fig. 5 | Schematic eigenvalues of the generator of the Kolmogorov operator, with indication of the spectral gap. In the isolated spectrum, the blue dots indicate the real eigenvalues and the black dots indicate the complex conjugate pairs. The continuous spectrum is indicated in grey. Tipping behaviour occurs when, as the bifurcation parameter approaches its critical value, the spectral gap $\tau = |\mathrm{Re}(\lambda_1)|$ shrinks to zero, in which λ_1 is the eigenvalue with smallest real part.

As discussed earlier, most of the early-warning-signal approaches outlined in this Review are essentially local, as they apply only in the vicinity of the tipping point. An alternate approach[140–142], based on landscape-flux theory[143], decomposes system dynamics into potential and flux components, thereby complementing the concept of a pseudopotential. Although this approach currently relies primarily on analyses of model equations, it could, in principle, be applied to time-series data to predict tipping points.

In conclusion to this section, we mention that the existing approaches to EWSs almost exclusively focus on B-tipping mechanism in the presence of small Gaussian noise. Correspondingly, the literature on EWS is dominated by discussion of the CSD as the main indicator of approaching tipping point. However, the CSD is not universally applicable[144] and there are a number of reasons for that: one of them being that CSD is designed for B-tipping but not necessarily for other tipping mechanisms[145–147]. Extending the theory of EWS onto other types of tipping — in particular, for AN-tipping, P-tipping and LT-tipping (Table 1) — is perhaps the biggest current challenge for anticipating critical transitions.

Case studies

The advanced theory of tipping points is a useful approach for understanding various ecological systems[1] and we illustrate this by focusing on three separate cases: kelp forest ecosystems[148–153], coral–algal–grazer systems[5,142,154–156] and savanna–forest systems[11,157,158]. In all these examples, the spatially local dynamics consists of transitions among states dominated by different species, with dispersal among different locations also potentially having a role. Thus, both the non-spatial versions of tipping and the spatial versions show up in understanding these systems on a global scale.

We begin by noting that any sufficiently large environmental change inevitably leads to ecosystems tipping in a cascade-like fashion which can be thought of as a leader–follower chain. One of the best-known cases addresses the importance of ocean currents for the marine systems, such as related to the AMOC. An AMOC collapse[159] is expected to strongly affect marine ecosystems, not only locally due to the change in water temperature and the disruption of nutrient supply[160] but also globally, through the strong AMOC feedback on global climate[161]. It is therefore important to understand the potential for the AMOC tipping (collapse), in particular by focusing on warning signals for the tipping[162], for example, based on an analysis extending ideas in refs. 146,163. Studies on the observed AMOC weakening (and hence its anticipated tipping) also emphasize the importance of statistical approaches[162].

Kelp forests, at least for centuries driven by human activities and changing climatic conditions, have undergone transitions between desirable states with kelp and predators, and undesirable states with sea urchin and calcareous algal-dominated kelp barrens, and kelp alone[148]. The transition from a state with kelp to a sea urchin barren is driven by overgrazing, which can be precipitated by a decline in predators of sea urchins and by climate change[148,152,153]. The presence of 'incipient barrens', which is essentially a spatial EWS of larger areas without kelp, provides a clear indication of likely tipping. For kelp systems, the potentially positive news is that both experiments[151] and an analysis focused on optimizing the approach[164] indicate that restoration of kelp forests is helped by management approaches that essentially consist of changing 'parameters' and the state by supporting predation and removing sea urchins.

Similar ideas arise in the study of coral–algal–grazer systems, in which grazing maintains the system in the desirable coral-dominated state. Tipping may occur when grazing pressure becomes insufficient, potentially because of declines in grazer populations[5,165]. As in the kelp example, there is good evidence of different final states, both from degradation at single sites and from spatial evidence[166]. More recent models[167] have examined how stochasticity interacts with the potential for tipping, and how transient dynamics near the tipping point can inform management strategies to promote the recovery and resilience of coral reefs in the face of challenges posed by the combined effects of climate change and anthropogenic stressors[6,7,168–170].

A terrestrial example involving similar dynamics is the transition between forest and savanna, which has been studied both empirically and through modelling. In this system, fire has a key role, much as grazing does in the preceding examples. As in those cases, both locally based dynamical approaches[11,171] and spatially explicit models[172,173] have been employed. Human activities that either suppress or increase fire, and climate change (which can increase fire), have an important role in governing the transition or tipping[174]. The ability to generate observed patterns from simple models[158] suggests that the models can indeed provide insights into management approaches as well as in understanding the forces governing current vegetation patterns and the effect that changing climate alone will have. For both the coral–algal–grazer system and the forest–savanna system, the potential-flux approach has been applied to the model equations[142,175].

Outlook

Despite the progress made over the past decade in understanding and predicting tipping points[176], in particular in ecosystems, many open questions remain and new challenges are emerging. First, ecosystems do not exist in isolation. All major systems on Earth are coupled. This applies not only to the climate-biosphere system, in which coupling between biotic and abiotic components is widely accepted[83,177,178], but also to somewhat less appreciated, more general, coupled socio-economical-ecological system[179,180]. There is growing evidence that regime shifts in social and/or economic systems can generate feedbacks affecting the global climate and environment[102,181]. In particular, shifts in human perceptions of ongoing climate and ecological crises, together with the resulting changes in behaviour — including attitudes towards the consumption of goods and dietary choices, as well as broader lifestyle changes — could substantially slow global environmental change[182]. Understanding of tipping scenarios in coupled socio-ecological-climatic system is a major challenge for future research.

Second, unlike abiotic systems, ecological systems exhibit remarkably high variability at all levels of organization. For example, individuals of the same species are not identical and hence can have different tolerance to a change in their ambient environment. Intuitively, owing to such variability and adaptivity, the system response to a change can be gradual rather than abrupt and hence may avoid tipping. However, a good understanding of that is lacking.

Third, many models of biological and ecological systems are known to show structural sensitivity to parametrization of interspecific interactions, in which implementing two different functions with very close shapes in a model would result in distinctly different model predictions, observed across a wide range of parameters[183–185]. How such structural sensitivity can affect models' capacity to describe tipping is an entirely open question.

Fourth, in this Review, our main focus is on theory and the corresponding mathematical models. Indeed, most studies of tipping points in ecosystems are model-based rather than data-driven, partly because

data of sufficient length and quality are rarely available. Moreover, ecosystems are often subject to substantial noise, which can make tipping points difficult to detect and even more challenging to anticipate[186]. An extensive meta-analysis of empirical data[187] showed that threshold-type ecosystem responses to a minor variation in forcing were rarely detectable. The observed responses were more typically characterized by gradual changes rather than abrupt ones. Qualitatively, this finding may be explained by the prevalence of partial typing[31–34] in real-world systems (which, given their great variability, can be intuitively expected). However, a quantitative investigation into this issue is missing, which poses another big challenge for future research.

We also note that the situation regarding ecological data has changed substantially over the past decade, with 'big data' becoming available for many ecosystems[188–190]. Although data quality remains an issue, the growing availability of such data creates opportunities for the broader application of data-driven methods, which could, in principle, provide a powerful complement — or alternative — to model-based approaches[136,137]. This is particularly true for studies on ecosystem dynamics in which it is hardly possible to include all the ecological complexity in the model, and hence the model construction can be subjective.

Understanding of tipping mechanisms and scenarios and the corresponding EWSs that accommodate the aforementioned challenges should be the overall direction of future research.

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Competing interests

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